

ONEDRAFT

CAD-AI — Aria Powered

Global Geospatial Application, Repository & World-Model Implementation Architecture

Version 1.0

Document: OneDraft Global Geospatial Application, Repository & World-Model Implementation Specification

System: OneDraft CAD-AI

Intelligence Layer: Aria

Document Class: Architectural & Implementation Stack Specification

Status: Architectural Specification

Predecessor: Global Geospatial Intelligence, Mapping & Earth Data Integration Architecture

Successor: OneDraft Global Spatial Data Processing & Distributed Repository Architecture

1. PURPOSE

Stack 52.0 defines the application architecture required to implement the Global Geospatial Intelligence, Mapping & Earth Data Integration Architecture established by Stack 51.0.

Stack 51.0 establishes the geographic architecture.

Stack 52.0 establishes how that architecture becomes an operational OneDraft application capability.

The system shall therefore progress from:

```
51.0
GLOBAL GEOSPATIAL ARCHITECTURE
    ↓
52.0
GLOBAL GEOSPATIAL APPLICATION
    ↓
SPATIAL REPOSITORY
    ↓
WORLD MODEL
    ↓
ONEDRAFT PROJECT INTEGRATION
```

Stack 52.0 establishes the application-layer implementation through which geographic information becomes accessible to OneDraft projects, Aria, the project model, the CAD/BIM environment, the documentation system and future global Earth-data systems.

2. SYSTEM OBJECTIVE

The objective of Stack 52.0 is to transform the geographic architecture established in Stack 51.0 into a persistent application capability capable of:

Registering geographic locations

Registering existing structures

Georeferencing OneDraft projects

Connecting projects to geographic context

Connecting existing-world information to projects

Maintaining spatial relationships

Maintaining temporal states

Maintaining source provenance

Synchronizing authorized GIS information

Supporting geographic search

Supporting spatial analysis

Supporting Aria geospatial operations

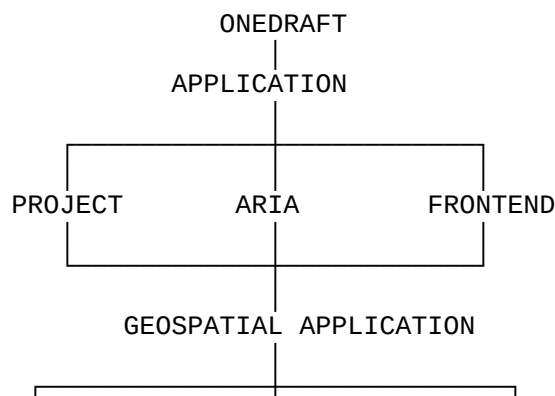
Maintaining a global project index

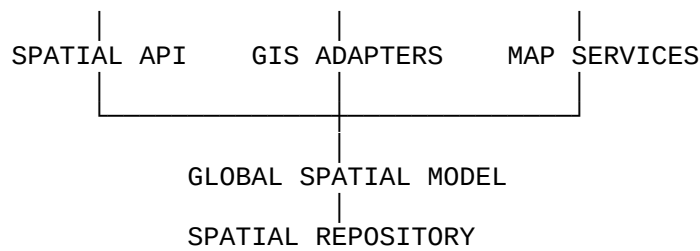
Providing a foundation for future Earth-data applications

The application shall be capable of expanding from individual project locations to a global spatial repository without requiring architectural replacement.

3. APPLICATION ARCHITECTURE

The Stack 52.0 application architecture shall integrate with the existing OneDraft platform as follows:





The geospatial application shall remain a bounded subsystem within the larger OneDraft architecture.

4. GLOBAL SPATIAL APPLICATION LAYER

The Global Spatial Application Layer shall provide the operational interface between the OneDraft project system and the geographic repository.

Its responsibilities shall include:

Spatial entity management

Geographic search

Spatial relationship management

GIS synchronization

Georeferencing

Spatial validation

Spatial metadata

Spatial provenance

Temporal spatial state

Map generation

Spatial analysis

Spatial project discovery

The application layer shall not directly bypass domain, authorization or persistence boundaries.

5. SPATIAL DOMAIN SERVICES

The application shall implement dedicated geospatial domain services.

Core services shall include:

`SpatialEntityService`

ProjectGeoreferenceService
SiteService
StructureService
SpatialRelationshipService
SpatialDatasetService
SpatialProvenanceService
SpatialVersionService
SpatialSearchService
SpatialValidationService
GISIntegrationService
MapService
SpatialAnalysisService

These services shall integrate with the existing OneDraft domain-service architecture.

6. PROJECT GEOREFERENCE SERVICE

The Project Georeference Service shall establish the geographic identity of a OneDraft project.

The service shall support:

Project coordinates

Site boundary

Building footprint

Coordinate reference system

Survey reference

Geographic identifiers

Jurisdictional identifiers

Elevation

Orientation

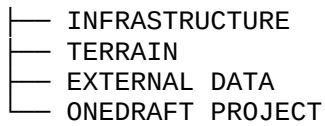
The georeference shall become part of the project domain state.

7. SITE SERVICE

The Site Service shall provide a canonical representation of the geographic project site.

A site shall be capable of containing:

SITE
|— LOCATION
|— BOUNDARY
|— EXISTING STRUCTURES



The site shall become the principal geographic bridge between external spatial information and the architectural project.

8. STRUCTURE SERVICE

The Structure Service shall maintain spatially indexed representations of existing and OneDraft-managed structures.

A structure record may contain:

Structure ID

Geographic location

Footprint

Height

Levels

Structure classification

Source

Temporal state

Project relationship

Provenance

Confidence

The structure record shall remain distinct from the complete architectural project model unless the two are explicitly linked.

9. GLOBAL PROJECT REGISTRY

The Global Project Registry shall provide the spatial index of OneDraft projects.

Conceptually:

GLOBAL
↓
CONTINENT
↓

COUNTRY
↓
REGION
↓
CITY
↓
SITE
↓
PROJECT

Every project possessing sufficient geographic information may receive a global spatial registry record.
The registry shall support authorized geographic discovery.

10. GLOBAL STRUCTURE REGISTRY

The Global Structure Registry shall provide a spatial index for known structures.

The registry may contain:

Existing structures

Historical structures

OneDraft projects

Proposed structures

Infrastructure structures

Other authorized built-environment records

The registry shall distinguish between source classifications.

11. WORLD-MODEL APPLICATION

The World-Model Application shall provide the persistent relationship between geographic entities and time.

The model shall support:

PAST
↓
HISTORICAL
↓
EXISTING
↓
CURRENT
↓
PROPOSED
↓

FUTURE

A spatial object shall therefore be capable of possessing multiple temporal states without requiring the destruction of earlier states.

12. SPATIAL ENTITY MODEL

The canonical spatial entity shall provide a common base for geographic records.

Conceptually:

```
SpatialEntity
├── id
├── geometry
├── spatial_reference
├── source
├── provenance
├── temporal_state
├── confidence
├── classification
└── metadata
```

Specialized entities shall inherit the common spatial architecture.

13. GEOGRAPHIC HIERARCHY SERVICE

The application shall maintain geographic hierarchy relationships.

The hierarchy shall support:

Planet

Continent

Country

Administrative region

Municipality

City

District

Site

Structure

The implementation shall accommodate jurisdictional differences rather than assuming a universal administrative model.

14. SPATIAL RELATIONSHIP ENGINE

The Spatial Relationship Engine shall calculate and persist authorized relationships.

Supported relationships shall include:

Contains

Contained By

Intersects

Overlaps

Adjacent To

Within

Near

Connected To

Located At

Derived From

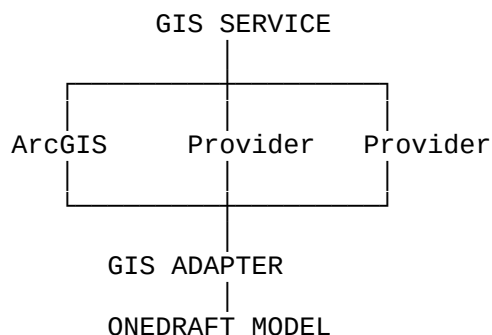
Modified From

Replaces

Relationships shall be validated before becoming persistent domain state.

15. GIS CONNECTOR ARCHITECTURE

The application shall provide a provider-neutral GIS connector architecture.



ArcGIS shall be implemented as a primary connector within this architecture.

Additional GIS systems may be integrated without changing the canonical OneDraft spatial model.

16. ARCGIS APPLICATION CONNECTOR

The ArcGIS Connector shall provide controlled interaction between OneDraft and authorized ArcGIS resources.

The connector shall support applicable capabilities including:

Layer discovery

Feature retrieval

Spatial queries

Feature metadata

Geographic attributes

Spatial references

Authorized synchronization

The connector shall not automatically treat external ArcGIS data as authoritative OneDraft project data.

17. GIS DATASET REGISTRY

The application shall maintain a registry of external geographic datasets.

A dataset record shall include:

Dataset

- Provider
- Source
- Dataset ID
- Layer
- Spatial Reference
- Version
- Acquisition Date
- License
- Access Classification
- Provenance

The registry shall allow OneDraft to understand where geographic information originated.

18. SPATIAL INGESTION SERVICE

The Spatial Ingestion Service shall process external geographic information.

The processing pipeline shall be:

SOURCE
↓
DISCOVERY
↓
RETRIEVAL
↓
SCHEMA INSPECTION
↓
SPATIAL REFERENCE IDENTIFICATION
↓
VALIDATION
↓
NORMALIZATION
↓
PROVENANCE
↓
CANONICAL MODEL
↓
INDEXING

Failed ingestion shall produce explicit errors and shall not silently modify canonical state.

19. SPATIAL NORMALIZATION SERVICE

The normalization service shall convert external geographic information into the OneDraft canonical spatial representation.

Normalization shall address:

Coordinate systems

Units

Geometry types

Attribute schemas

Identifiers

Temporal representations

Classification systems

The original source representation shall remain recoverable where required.

20. COORDINATE TRANSFORMATION SERVICE

The Coordinate Transformation Service shall perform controlled spatial-reference transformations.

Every transformation shall preserve:

Source CRS

Target CRS

Transformation method

Units

Precision

Transformation metadata

The transformed geometry shall remain traceable to the original representation.

21. SPATIAL VALIDATION SERVICE

The Spatial Validation Service shall verify incoming and generated spatial information.

Validation shall include:

Geometry validity

Coordinate validity

Spatial reference

Topology

Required attributes

Boundary consistency

Precision

Schema compatibility

The validation result shall be recorded.

22. SPATIAL VERSION SERVICE

The Spatial Version Service shall maintain geographic versions.

A spatial entity may therefore progress:

VERSION 1

↓

VERSION 2

↓

VERSION 3

↓

CURRENT

Historical states shall remain identifiable according to retention policy.

23. WORLD-STATE SERVICE

The World-State Service shall provide temporal state management.

It shall distinguish:

Historical

Existing

Current

Proposed

Future

The service shall permit comparison between states.

24. CHANGE REGISTRY

The Change Registry shall record authorized spatial changes.

Examples include:

Construction

Alteration

Expansion

Demolition

Replacement

Boundary Change

Infrastructure Change

Project Proposal

Changes shall be associated with time, source and provenance.

25. GLOBAL DATA PLACEHOLDER SYSTEM

One of the primary functions of Stack 52.0 shall be the creation of persistent geographic placeholders.

A geographic record may exist before complete information is available.

For example:

LOCATION
↓
PLACEHOLDER RECORD
↓
STRUCTURE IDENTIFIED
↓
ADDITIONAL DATA
↓
ONEDRAFT PROJECT
↓
ENGINEERING INFORMATION
↓
DOCUMENTATION

The placeholder system permits the repository to grow incrementally.

26. EXISTING-WORLD IMPORT

Existing-world information may be incorporated into the repository where authorized and available.

Potential sources include:

GIS

CAD

BIM

Survey

Public geographic information

Licensed datasets

Client data

Historical records

Each source shall retain provenance.

27. ONEDRAFT PROJECT CONTRIBUTION

A completed or appropriately validated OneDraft project may contribute authorized spatial information back into the global repository.

The contribution workflow shall be:

```
PROJECT
↓
VALIDATION
↓
ACCEPTED PROJECT STATE
↓
GEOGRAPHIC EXTRACTION
↓
PROVENANCE
↓
GLOBAL REGISTRY
```

The system shall not automatically publish private project information.

Publication shall be governed by authorization and data policy.

28. OLD + NEW DATA CONVERGENCE

The application shall support convergence between pre-existing information and newly generated OneDraft information.

```
HISTORICAL DATA
↓
GLOBAL SPATIAL MODEL
↑
NEW ONEDRAFT PROJECT
```

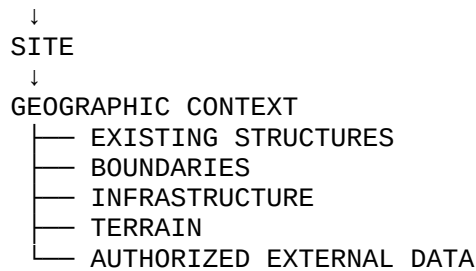
Both may become temporally related records within the same geographic environment.

29. ARCHITECTURAL CONTEXT SERVICE

The Architectural Context Service shall allow OneDraft to retrieve relevant geographic information for a project.

A project request may produce:

```
PROJECT
```

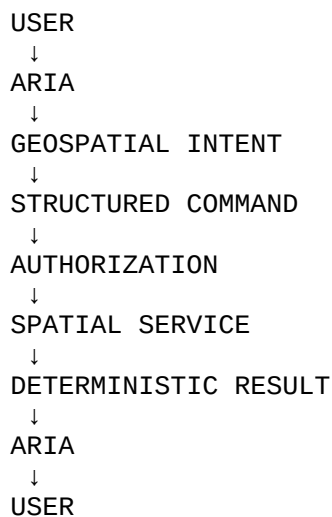


This context may then be supplied to project workflows.

30. ARIA GEOSPATIAL SERVICE

Aria shall interact with the Global Geospatial Application through controlled service interfaces.

The flow shall be:



Aria shall not directly modify canonical spatial records without the appropriate service and authorization path.

31. SPATIAL QUERY SERVICE

The Spatial Query Service shall support:

Radius searches

Bounding-box searches

Polygon searches

Intersection queries

Containment queries

Nearest-feature queries

Geographic hierarchy queries

Project-location queries

Temporal queries

Results shall remain subject to authorization.

32. GLOBAL MAP APPLICATION

The frontend shall provide a geographic visualization layer.

The conceptual navigation shall be:

PLANET
↓
CONTINENT
↓
COUNTRY
↓
CITY
↓
SITE
↓
STRUCTURE
↓
PROJECT
↓
DOCUMENTATION

Map visualization shall be a representation of underlying spatial data and shall not become the authoritative source of project geometry.

33. PROJECT MAP CONTEXT

Every georeferenced OneDraft project shall be capable of displaying its geographic context.

The project map may include:

Site

Building

Adjacent structures

Roads

Boundaries

Infrastructure

Terrain

Authorized external layers

The map shall remain synchronized with the project spatial state.

34. CAD / GIS APPLICATION BRIDGE

The application shall provide controlled translation between architectural geometry and geographic geometry.

```
CAD / BIM
  ↓
PROJECT COORDINATE SYSTEM
  ↓
GEOREFERENCE
  ↓
GIS REPRESENTATION
```

and:

```
GIS
  ↓
SITE CONTEXT
  ↓
PROJECT COORDINATE SYSTEM
  ↓
CAD / BIM CONTEXT
```

Transformations shall preserve source and precision metadata.

35. GLOBAL STRUCTURE PLACEHOLDERS

Where an existing structure is identified geographically but detailed architectural information is unavailable, the application may maintain a structure placeholder.

For example:

```
STRUCTURE ID
LOCATION
FOOTPRINT
SOURCE
DATE
STATUS
```

Later information may populate:

BUILDING MODEL
LEVELS
USE
MATERIALS
DOCUMENTATION
ALTERATIONS
ENGINEERING DATA

The placeholder architecture is therefore a direct mechanism for progressive repository development.

36. TEMPORAL PROJECT INTEGRATION

A OneDraft project shall be capable of being associated with multiple world states.

EXISTING SITE
↓
PROJECT DESIGN
↓
APPROVED DESIGN
↓
CONSTRUCTION
↓
COMPLETED STRUCTURE
↓
FUTURE ALTERATION

The geographic record shall remain connected throughout the lifecycle.

37. GLOBAL PROJECT GRAPH

The application shall support a geographic project graph.

Conceptually:

CITY
├── SITE
│ ├── EXISTING STRUCTURE
│ ├── ONEDRAFT PROJECT
│ └── INFRASTRUCTURE
├── SITE
│ └── PROJECT
└── SITE
 └── STRUCTURE

This graph shall permit spatial and relational discovery.

38. SPATIAL KNOWLEDGE GRAPH FOUNDATION

The geographic application shall establish the foundation for a future spatial knowledge graph.

Relationships may connect:

PLACE
↓
STRUCTURE
↓
PROJECT
↓
DOCUMENT
↓
EVENT
↓
TIME
↓
SOURCE

This creates machine-readable relationships across geographic and engineering information.

39. GLOBAL REPOSITORY WRITE-BACK

Where authorized, OneDraft shall write validated project-derived information back into the global repository.

Write-back shall occur only after:

PROJECT VALIDATION
↓
DATA CLASSIFICATION
↓
AUTHORIZATION
↓
PROVENANCE
↓
REPOSITORY UPDATE

This prevents incomplete or unauthorized project information from becoming global repository state.

40. GLOBAL REPOSITORY READ-BACK

Existing global information may be read into a project context through:

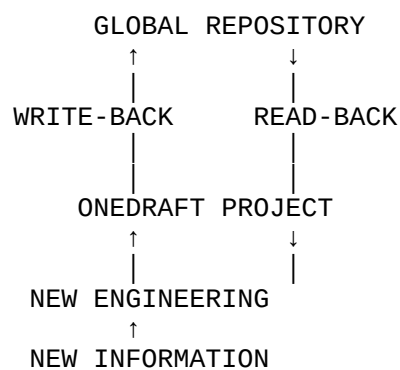
GLOBAL REPOSITORY
↓

SPATIAL QUERY
↓
DATA VALIDATION
↓
PROJECT CONTEXT
↓
ONEDRAFT

Read-back data shall remain distinguishable from project-generated information.

41. BIDIRECTIONAL INFORMATION LOOP

The complete application architecture establishes:



This creates the foundation for a continuously expanding geographic engineering repository.

42. DATA PROVENANCE APPLICATION

Every significant spatial record shall retain provenance through the application layer.

The application shall preserve:

Origin

Source

Transformation

Version

Timestamp

Processing

Project

Operator or system process

Provenance shall remain available to authorized users and system services.

43. DATA QUALITY APPLICATION

The application shall maintain quality metadata.

Quality dimensions may include:

Accuracy

Precision

Completeness

Currency

Resolution

Confidence

Verification State

Quality information shall remain distinct from the underlying geographic record.

44. SPATIAL CONFIDENCE

Spatial information may be classified as:

VERIFIED
SOURCE-CONFIRMED
IMPORTED
DERIVED
ESTIMATED
UNRESOLVED
SUPERSEDED

The classification shall communicate the state of the information without converting uncertainty into false precision.

45. GLOBAL SEARCH APPLICATION

The application shall provide geographic discovery across authorized records.

Users may search by:

Location

Project

Structure

City

Country

Dataset

Time

Project status

Structure type

Spatial relationship

Search results shall respect tenant and data-access boundaries.

46. SPATIAL ANALYSIS SERVICE

The application shall expose deterministic spatial operations including:

Distance

Area

Intersection

Containment

Proximity

Buffering

Spatial aggregation

Coordinate transformation

The resulting values shall be computational outputs rather than AI-generated estimates.

47. ARIA ANALYTICAL ORCHESTRATION

Aria may orchestrate spatial analysis.

For example:

USER:

"Show the relevant existing structures around this site."

ARIA
↓
INTERPRET REQUEST
↓
SPATIAL QUERY
↓
DETERMINISTIC FILTER
↓
RESULT
↓
CONTEXTUAL RESPONSE

The system shall preserve the distinction between computation and interpretation.

48. FUTURE EARTH-DATA APPLICATION INTERFACE

Stack 52.0 shall establish application interfaces capable of supporting future datasets beyond architecture.

Potential future domains include:

Ecology

Land

Water

Climate

Infrastructure

Transportation

Human settlement

Humanitarian mapping

Population movement

Environmental monitoring

These systems shall integrate through the Global Spatial Application rather than directly modifying architectural project services.

49. MASS-DISPLACEMENT APPLICATION FOUNDATION

The spatial application shall be capable of supporting future aggregate movement datasets.

The conceptual model shall be:

ORIGIN
↓
MOVEMENT EVENT
↓
GEOGRAPHIC PATH
↓
TEMPORARY LOCATION
↓
DESTINATION

Where such systems are later implemented, privacy, aggregation and data-governance controls shall apply.

50. EARTH-ECOLOGY APPLICATION FOUNDATION

The application shall provide a geographic interface through which future ecological information may be spatially associated with built-environment information.

BUILT ENVIRONMENT
+
LAND
+
WATER
+
ECOLOGY
+
TIME
↓
SPATIAL EARTH MODEL

The implementation shall remain modular.

51. GLOBAL WORLD-MODEL APPLICATION

The World-Model Application shall provide the central relationship:

WHERE
+
WHAT
+
WHEN
+
SOURCE
+
CHANGE

for geographically represented information.

This creates the foundation for a continuously evolving spatial world model.

52. REPOSITORY MATURATION

The repository shall mature progressively.

The intended progression is:

PLACEHOLDER
↓
LOCATION
↓
STRUCTURE
↓
CONTEXT
↓
PROJECT
↓
ENGINEERING
↓
DOCUMENTATION
↓
VERIFIED CHANGE
↓
HISTORICAL RECORD

A geographic record therefore does not need to be complete at creation.

It becomes progressively richer.

53. CITY-TO-CITY EXPANSION

The application shall support expansion from individual projects to city-scale datasets.

PROJECT
↓
CITY INDEX
↓
MULTIPLE PROJECTS

↓
STRUCTURE REGISTRY
↓
CITY BUILT-ENVIRONMENT MODEL

The same architecture shall then extend to regions and countries.

54. COUNTRY-TO-COUNTRY EXPANSION

The repository shall support geographic expansion:

CITY
↓
REGION
↓
COUNTRY
↓
COUNTRY
↓
COUNTRY

The application architecture shall remain jurisdiction-neutral while permitting jurisdiction-specific geographic and regulatory configuration.

55. CONTINENTAL EXPANSION

The system shall support:

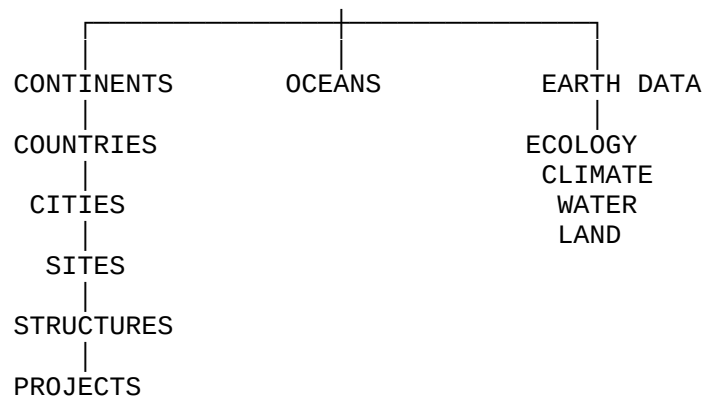
COUNTRY
↓
CONTINENT
↓
MULTI-CONTINENT
↓
GLOBAL

Spatial indexing and infrastructure shall scale independently of the project-domain architecture.

56. PLANETARY APPLICATION MODEL

The long-term application model shall become:

PLANET
|
GLOBAL SPATIAL INDEX
|



The planetary model shall be understood as an architectural target enabled by progressive repository growth.

57. API INTEGRATION

Stack 52.0 shall integrate with the existing OneDraft API architecture.

Spatial APIs shall support:

```
/projects/{id}/location  
/sites/{id}  
/structures/{id}  
/spatial/entities  
/spatial/datasets  
/spatial/search  
/spatial/relationships  
/spatial/versions  
/spatial/events
```

Final API contracts shall be governed through the existing OpenAPI architecture.

58. EVENT INTEGRATION

Spatial events shall enter the OneDraft Event Bus.

Events may include:

```
ProjectGeoreferenced  
SiteCreated  
StructureRegistered  
SpatialDatasetImported  
SpatialDatasetUpdated  
SpatialRelationshipCreated  
SpatialStateChanged  
GISSynchronizationCompleted  
SpatialValidationCompleted  
GlobalRepositoryUpdated
```

Events shall be versioned and traceable.

59. JOB INTEGRATION

Large geospatial operations shall execute through the existing Job & Worker system.

Jobs may include:

GIS imports

Spatial transformations

Large-area searches

Spatial indexing

Map rendering

Dataset synchronization

Spatial analysis

Repository reconciliation

The application shall not execute long-running operations synchronously within ordinary request paths.

60. ARTIFACT INTEGRATION

Generated geographic outputs shall use the existing Artifact Management Engine.

Artifacts may include:

Maps

Spatial reports

GeoJSON

CAD exports

GIS packages

Spatial datasets

Rendered geographic context

Artifacts shall receive hashes, provenance and lifecycle metadata.

61. SECURITY INTEGRATION

All spatial operations shall pass through the existing OneDraft security architecture.

The application shall enforce:

Authentication

Authorization

Tenant isolation

Project permissions

Dataset permissions

Administrative privileges

Audit logging

No global geographic index shall bypass private-project access controls.

62. MULTI-TENANT SPATIAL MODEL

Spatial information shall be classified according to visibility:

GLOBAL / PUBLIC
|
LICENSED
|
ORGANIZATION
|
PROJECT
|
RESTRICTED

The same geographic location may therefore contain information belonging to different access domains.

The application shall enforce those boundaries independently of geographic proximity.

63. ADMINISTRATIVE CONTROL

Stack 49.0 shall control application-level geographic configuration.

Administrative controls may include:

GIS providers

Dataset sources

Spatial policies

Coordinate systems

Import rules

Data classifications

Repository write-back rules

Geospatial feature availability

Administrative changes shall be versioned and audited.

64. OBSERVABILITY

Stack 52.0 shall integrate with OneDraft telemetry.

Metrics shall include:

Spatial query latency

GIS connector availability

Dataset ingestion rate

Transformation failures

Spatial validation failures

Repository writes

Repository reads

Spatial job duration

Map rendering performance

The system shall provide operational visibility into the global geospatial application.

65. TESTING

The geospatial application shall require:

Unit testing

Integration testing

GIS connector testing

Spatial geometry testing
Coordinate transformation testing
API contract testing
Security testing
Tenant-isolation testing
Temporal-state testing
Provenance testing
Performance testing
End-to-end project georeferencing

66. FAILURE HANDLING

The application shall explicitly handle:

GIS unavailable
Dataset unavailable
Invalid geometry
Invalid coordinates
Unsupported CRS
Transformation failure
Schema mismatch
Authorization failure
Duplicate record
Conflicting source
Repository failure

Failure shall produce recoverable state wherever practical.

67. SYNCHRONIZATION MODEL

Synchronization shall support controlled states:

DISCOVERED

↓
REGISTERED
↓
SYNCHRONIZING
↓
VALIDATING
↓
SYNCHRONIZED
↓
SUPERSEDED

The system shall not assume that external sources remain static.

68. CONFLICT MODEL

When external GIS information differs from existing OneDraft information, the application shall create a conflict state where appropriate.

SOURCE A
 \
 → CONFLICT → RESOLUTION
 /
SOURCE B

Resolution shall be governed by configured rules and, where necessary, human review.

69. DATA LINEAGE

The application shall maintain lineage:

EXTERNAL SOURCE
↓
DATASET
↓
FEATURE
↓
TRANSFORMATION
↓
ONEDRAFT ENTITY
↓
PROJECT
↓
ARTIFACT

This permits downstream information to remain traceable.

70. APPLICATION LIFECYCLE

The Stack 52.0 application lifecycle shall be:

```
REGISTER
↓
LOCATE
↓
INGEST
↓
NORMALIZE
↓
VALIDATE
↓
INDEX
↓
CONNECT
↓
ANALYZE
↓
UPDATE
↓
VERSION
↓
REUSE
```

This lifecycle shall operate continuously as the repository expands.

71. ENGINEERING INFORMATION CONTINUITY

The application establishes continuity between previous and future engineering work.

```
PREVIOUS PROJECT
↓
GLOBAL REPOSITORY
↓
CURRENT PROJECT
↓
NEW ENGINEERING
↓
GLOBAL REPOSITORY
↓
FUTURE PROJECT
```

This is the operational implementation of the architectural principle established in Stack 51.0.

72. PROJECT-TO-REPOSITORY FEEDBACK

A completed project can become a source of contextual information for future projects.

The controlled loop is:

```
PROJECT
↓
VALIDATED INFORMATION
↓
AUTHORIZED CONTRIBUTION
↓
GLOBAL REPOSITORY
↓
FUTURE PROJECT
```

This creates cumulative information growth.

73. REPOSITORY-TO-PROJECT FEEDBACK

The reverse process is:

```
GLOBAL REPOSITORY
↓
AUTHORIZED DATA
↓
PROJECT CONTEXT
↓
ARIA
↓
DESIGN / ENGINEERING
```

The two directions establish a persistent information cycle.

74. CONTINUOUS ENGINEERING MODEL

The complete Stack 52.0 implementation produces:

```
OBSERVE
↓
REGISTER
↓
MAP
↓
MODEL
↓
ENGINEER
↓
VALIDATE
↓
```

DOCUMENT
↓
BUILD / MODIFY
↓
RECORD
↓
UPDATE
↓
OBSERVE AGAIN

Engineering information therefore becomes cumulative rather than strictly project-isolated.

75. ARCHITECTURAL INVARIANTS

Stack 52.0 shall preserve the following invariants:

Geographic Identity: Every georeferenced project has a persistent spatial identity.

Provenance: External information remains attributable to its source.

Temporal Integrity: Historical states remain distinguishable.

Authority: External GIS information does not automatically override OneDraft project state.

Determinism: Spatial calculations use deterministic services.

AI Separation: Aria interprets and orchestrates but does not replace authoritative spatial computation.

Security: Geographic location does not override access control.

Scalability: The application can grow from project scale to planetary scale.

Interoperability: GIS, CAD, BIM and OneDraft remain connected through defined interfaces.

Extensibility: Future Earth-data systems can attach without restructuring the core architecture.

76. END-TO-END APPLICATION WORKFLOW

The complete Stack 52.0 workflow is:

USER
↓
ONEDRAFT PROJECT
↓
SITE IDENTIFICATION
↓
GEOREFERENCE
↓
GLOBAL SPATIAL QUERY

↓
ARCGIS / GIS DATA
↓
INGESTION
↓
NORMALIZATION
↓
VALIDATION
↓
GLOBAL SPATIAL CONTEXT
↓
PROJECT MODEL
↓
ARIA
↓
ENGINEERING / DESIGN
↓
VALIDATION
↓
DOCUMENTATION
↓
ACCEPTANCE
↓
AUTHORIZED REPOSITORY WRITE-BACK
↓
GLOBAL WORLD MODEL

This workflow establishes the operational connection between the OneDraft project and the geographic world.

77. SYSTEM INTEGRATION WITH 1.0–51.0

The architecture now becomes:

1.0–5.0
FOUNDATION
↓
6.0–16.0
EXECUTABLE APPLICATION FOUNDATION
↓
17.0–22.0
CORE COMPUTATIONAL & DOCUMENTATION ENGINES
↓
23.0–29.0
RUNTIME, JOBS, ARTIFACTS, EVENTS & SECURITY
↓
30.0–36.0
OPERATIONS, QUALITY, INFRASTRUCTURE & DATA GOVERNANCE
↓
37.0–40.0
AI, CAD STANDARDS & DATA EXCHANGE
↓
41.0–49.0
USER PLATFORM, LIFECYCLE, COMMERCIAL & SCALE SYSTEMS
↓

50.0
END-TO-END SYSTEM INTEGRATION
↓
51.0
GLOBAL GEOSPATIAL INTELLIGENCE & MAPPING ARCHITECTURE
↓
52.0
GLOBAL GEOSPATIAL APPLICATION & WORLD-MODEL IMPLEMENTATION

78. ARCHITECTURAL COMPLETION STATE

With Stack 52.0, OneDraft now contains the architecture necessary to connect individual architectural projects to a persistent global spatial information environment.

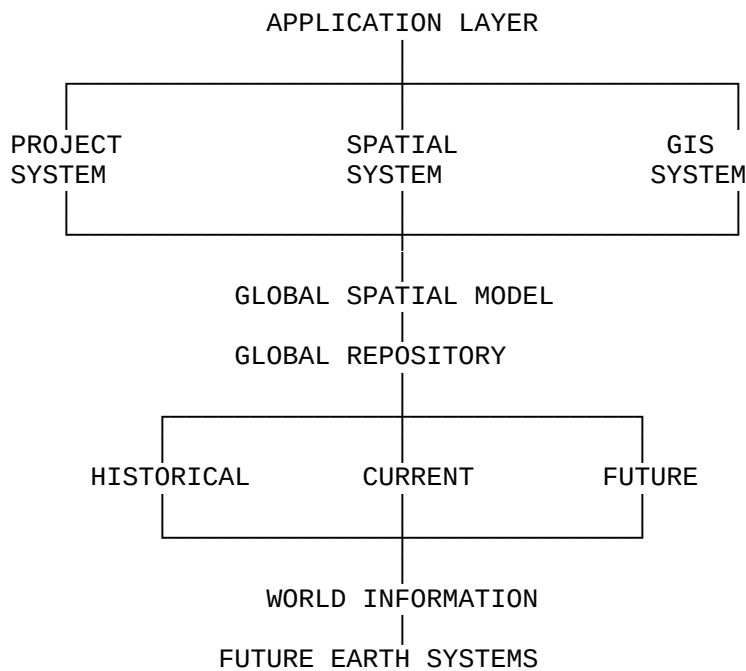
The resulting system is:

PROJECT
+
GEOMETRY
+
COMPLIANCE
+
VALIDATION
+
DOCUMENTATION
+
ARIA
+
GIS
+
GLOBAL SPATIAL REPOSITORY
+
TEMPORAL WORLD MODEL
+
EXISTING STRUCTURES
+
FUTURE STATES
↓
ONEDRAFT GLOBAL ENGINEERING PLATFORM

79. MASTER APPLICATION MODEL

The complete Stack 52.0 application can be represented as:

ONEDRAFT
|
ARIA
|



80. FINAL SYSTEM PRINCIPLE

Stack 52.0 establishes the operational principle:

THE WORLD PROVIDES THE CONTEXT.
GIS PROVIDES THE GEOGRAPHIC FRAMEWORK.
ONEDRAFT PROVIDES THE ENGINEERING MODEL.
ARIA PROVIDES THE INTELLIGENT INTERFACE.
VALIDATION PROVIDES DETERMINISTIC CONTROL.
THE REPOSITORY PROVIDES CONTINUITY.
TIME PROVIDES HISTORY.
NEW PROJECTS PROVIDE NEW INFORMATION.

The result is an architecture in which every authorized OneDraft project can become part of a larger, geographically indexed and temporally aware engineering information environment.

81. COMPLETION STATUS

Stack: 52.0

System: OneDraft CAD-AI

Intelligence Layer: Aria

Primary Function: Global Geospatial Application, Repository & World-Model Implementation

Primary Integration: GIS / ArcGIS

Primary Repository: OneDraft Global Spatial Repository

Primary Model: Global Spatial + Temporal World Model

Application Scope: Project → Site → Structure → City → Region → Country → Continent → Global

Temporal Scope: Historical → Existing → Current → Proposed → Future

Integration Scope: CAD + BIM + GIS + AI + Architectural Engineering + Global Spatial Data

Architectural Status: DEFINED

System Register: 1.0–52.0

The OneDraft architecture now progresses from the original architectural project system into a geographically persistent engineering information system capable of progressively connecting existing structures, new projects and future information through a common spatial and temporal framework.